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Self-oscillating synchronematic colloids

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Self-oscillators that sustain periodic dynamics under constant input are ubiquitous in natural and engineered systems, where their interactions enable spatiotemporal coordination among many individual units. New forms of organization can emerge when these self-oscillating units are free to move and rotate, linking their spatial arrangement and orientation with their oscillation frequencies and phases. Here, we report experiments and simulations on populations of Quincke colloids that behave as self-oscillating units characterized by position, orientation, frequency, and phase. Hydrodynamic interactions among these colloids drive temporal synchronization and spatial alignment of their phases and orientations, giving rise to a new form of collective order that we term synchronematic. Within finite-size crystalline clusters, these non-reciprocal interactions promote global synchronization and circular alignment, with a collective frequency that increases with cluster size. Using the theory of weakly coupled oscillators, we derive a reduced-order model that captures the coupled evolution of phase and orientation and explains how synchronematic order depends sensitively on the particle configuration. Our results establish Quincke colloids as a model system for active oscillatory matter and reveal fundamental principles by which synchronization, alignment, and structure co-emerge—offering a framework for designing adaptive, frequency-tunable materials.

Self-oscillators—dissipative systems that sustain periodic dynamics in response to steady inputs—are ubiquitous across physics, engineering, and biology, from quartz timekeepers to beating cilia[1]. Though diverse in form, these systems share a common feature: a phase variable that tracks progress through the oscillation cycle and mediates interactions, giving rise to collective behaviors such as synchronization[2,3], wave propagation[4,5], and pattern formation[6,7]. In the seminal Kuramoto model, oscillator phases influence one another through pairwise interactions that promote synchronization despite heterogeneity in their natural frequencies[6,8]. In the case of identical oscillators with local, reciprocal couplings, the Kuramoto model becomes mathematically equivalent to the XY model, a classical framework for describing orientational order in two-dimensional spin systems[9,10]. This equivalence highlights the close analogy between temporal phase alignment in oscillator ensembles and spatial orientation alignment—for example, in nematic liquid-crystal films[11]. In both cases, the interactions derive from gradients of a scalar potential, ensuring reciprocity and enabling analysis within the framework of equilibrium statistical mechanics.

Active matter goes beyond this equilibrium framework to describe collections of self-driven units that interact and evolve through non-conservative forces that do not derive from a global potential[12]. In theoretical models, such forces may arise from self-propulsion, as in active Brownian particles[13,14], or from non-reciprocal interactions, as in phoretic colloids that break action-reaction symmetry[15]. The resulting dynamics give rise to collective behaviors that depend on the symmetries of the constituent units and their interactions—whether isotropic or anisotropic (e.g., polar or nematic),

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chiral or achiral. Analogous effects appear in the theory of coupled oscillators: the Sakaguchi-Kuramoto model introduces non-reciprocal (and thereby non-conservative) couplings that shift the collective oscillation frequency and modify the onset of synchronization[16]. More recently, models of mobile oscillators with phase and position—so-called "swarmalators"—have bridged the domains of coupled oscillators and active matter by revealing new modes of organization that combine temporal synchronization and spatial assembly[17,18]. Although few experiments have explored aspects of these models[19,20], there remains a lack of experimental systems that manifest the rich behaviors that may arise when phase, position, and orientation are coupled—as in the flapping flight of birds.

Active colloids[21,22] based on Quincke rotation in a static electric field provide a versatile experimental platform for exploring collective dynamics in active matter[23–26]. In these systems, dielectric particles roll across the electrode surface due to charge accumulation at the particle-fluid interface and its mechanical relaxation in the external field[27,28]. These self-driven particles provide a useful model for polar active fluids[23] but can also realize distinct types of self-oscillatory motion—even in the absence of particle inertia[29]. In one case, asymmetric (e.g., pear-shaped) particles undergo circular Quincke rotation, tracing closed orbits to create chiral oscillators with time-averaged isotropic dynamics[30]. In another, spherical particles exhibit axial Quincke oscillations, rolling back and forth along a spontaneously selected axis[31]. These oscillations are achiral and time-averaged nematic, characterized by reflection symmetry in the plane and an orientation that lacks a head or tail. Unlike colloids driven by external time-periodic forcing[32–34], each Quincke oscillator has an independent frequency and phase that mediate its interactions with neighboring particles. These active units provide an opportunity to explore how phase synchronization and orientational alignment—long treated separately in the Kuramoto and XY models—combine and interact under nonequilibrium conditions.

Here, we investigate populations of self-oscillating colloids and identify distinct modes of collective organization that depend on the spatial distribution of the particles. We combine these experiments with Stokesian Dynamics (SD) simulations[35–37], which capture the particle dynamics and hydrodynamic couplings, to elucidate the origins of these collective modes. In fluid-like clusters with smooth particle distributions, oscillators develop local order in both phase and orientation, reflecting their tendency to synchronize and align. We quantify this synchronematic order and show that it emerges from the competition between hydrodynamic interactions and athermal noise[38]. In configurations with sharp density contrasts, particles spontaneously assemble into finite crystalline clusters exhibiting global synchronization and circular alignment within each cluster—structures we refer to as synchronematic crystals. These dynamic assemblies are reproduced in simulations that combine electroosmotic flows, which drive cluster densification, with hydrodynamic interactions, which mediate synchronization and alignment. Using the theory of weakly coupled oscillators, we develop a reduced-order, swarmalator-type model that describes the joint evolution of oscillator phases and orientations. The derived interactions include both Kuramoto-like variational and Sakaguchi-type non-variational terms, which explain the observed increase in oscillation frequency with particle density and cluster size. Together, these findings demonstrate that hydrodynamic interactions among self-oscillating colloids couple phase synchronization and orientational alignment, integrating and extending the Kuramoto and XY models into a unified framework that may prove useful in the design of active materials and robotic collectives.

Results

Experimental summary

Our experiments are based on collections of self-oscillating microspheres dispersed in a weakly conductive fluid above a planar electrode subject to a static electric field (Fig. 1a). In contrast to Quincke rollers[23], which move steadily across the electrode surface, the present conditions give rise to oscillatory dynamics, whereby individual particles roll back and forth along a spontaneously selected direction[31]. The periodic motion of each Quincke oscillator is characterized by the oscillation frequency ω, amplitude A, phase θ, and orientation χ as illustrated in Fig. 1b, c. We infer these variables from particle trajectories using a stochastic model that accounts for diffusive fluctuations in the oscillator phase, angle, and position ("Methods"; Supplementary Note 1). The particle dynamics exhibit a phase-angle symmetry, in which a shift in phase by π is equivalent to a 180° rotation of the orientation angle[39].

By varying the initial particle distribution, we observe three distinct modes of organization with different characteristic frequencies (Fig. 1d): (i) dilute non-interacting particles[31], (ii) a locally correlated fluid cluster (Fig. 1e), and (iii) oscillating clusters with crystalline structure (Fig. 1f). Under dilute conditions, particles oscillate independently with an average frequency $\omega = 1060 \pm 23$ rad/s (Fig. 1c), while their phase θ and orientation χ exhibit diffusive motion due to athermal noise associated with the field-driven oscillations. By analyzing the trajectories of different particles, we estimate the corresponding diffusivities as $D_\theta = 5.2 \pm 1.2$ rad^2/s and $D_\chi = 4.3 \pm 0.6$ rad^2/s (Supplementary Note 1). This noise will be shown to play a key role in determining the correlations among interacting oscillators. At higher area fractions, qualitatively different structures emerge depending on the initial particle distribution: initially uniform suspensions evolve into extended, fluid-like clusters (Fig. 1e), whereas suspensions seeded with locally dense regions develop compact crystalline assemblies (Fig. 1f). Aside from these differences in the initial distribution, the experimental conditions—particle size, field strength, electrolyte composition, and overall density—are the same in both cases (Methods).

Within the fluid cluster, interactions among oscillating particles lead to an increase in their average frequency (Fig. 1d), accompanied by local ordering in both phase and orientation. In the experiment shown in Fig. 1e, particles are initially positioned by repulsive dipolar interactions under a weak electric field that does not induce rolling or oscillation. Despite density variations on larger scales, particles within the imaging region (ca. 1 mm) adopt uniform interparticle spacing at an average density of $\rho \approx 0.1$. Upon activation of particle oscillations, the local density remains constant, and the system relaxes to a pseudo-steady state characterized by local synchronization and alignment of neighboring oscillators, as illustrated by the colored ellipsoidal markers (Fig. 1e; Supplementary Videos 1 and 3). The orientation of each ellipsoid represents the oscillation angle χ, while its color denotes the oscillation phase θ (modulo π).

Instead, when initialized within local regions of high density, particles self-assemble into discrete crystalline clusters that exhibit synchronized oscillations with circular alignment (Fig. 1f; Supplementary Video 2). We hypothesize that these clusters are stabilized by electroosmotic flows that promote local densification[40–44]. Confined by such flows, particles form close-packed assemblies that synchronize and align to perform collective vortex-like motions around a central topological defect. Each cluster behaves as a coherent unit, oscillating independently of neighboring clusters with different phases at frequencies approaching 1600 rad/s (Fig. 1d).

To explain these different modes of organization, we performed simulations that account for the intrinsic oscillatory dynamics of each particle[39] as well as the hydrodynamic, electrostatic, and steric interactions that couple them (Supplementary Note 2). To model hydrodynamic interactions at low Reynolds numbers, we adapt an SD method that incorporates far-field many-body interactions among particles above a no-slip wall and near-field lubrication interactions between nearly touching surfaces[37]. Electrostatic interactions are treated through a multipolar expansion, in which particles interact through the electric fields and forces generated by their dipole and

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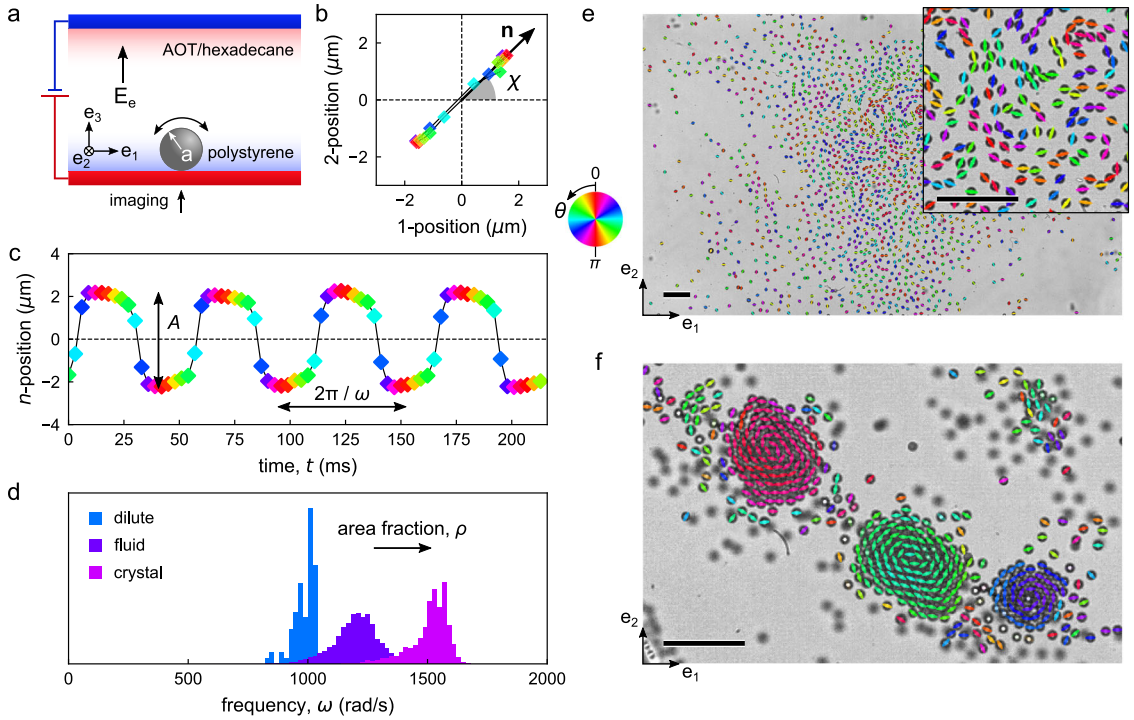


Fig. 1 | Self-oscillating Quincke colloids. a Schematic illustration of a Quincke oscillator of radius a. **b** Particle oscillations are characterized by the orientation angle χ measured from the 1-axis (or by the director **n**). Colored markers denote the measured particle position relative to the oscillation center at regular intervals. **c** The particle position projected along the **n**-direction oscillates with frequency ω and amplitude A. Marker colors denote the oscillation phase θ, which increases linearly in time as $\theta = \omega t + \phi$ with relative phase ϕ. **d** Histograms of the oscillation frequency ω in the dilute, fluid, and crystalline regimes suggest that ω increases with local particle density. **e** Image of a low-density fluid cluster annotated by ellipsoidal markers denoting the orientation and phase of each particle oscillator. The inset shows a magnified image of the cluster center. Scale bars are 100 μm. See also Supplementary Videos 1 and 3. **f** Crystalline assemblies of synchronized oscillators with circular alignment. Out-of-focus particles are oscillating on the upper electrode. Scale bar is 100 μm. See also Supplementary Video 2.

quadrupole moments (see "Methods"). At the single-particle level, the simulations quantitatively reproduce the velocity of rolling particles and the frequency of oscillating particles observed in experiments (Supplementary Fig. S4). Simulations of interacting pairs further demonstrate the tendency of neighboring oscillators to synchronize and align through their mutual interactions (Supplementary Note 3).

Fluid clusters

Using experimental and simulation data for the low-density fluid cluster, we compute pairwise statistics to quantify local ordering in the oscillator phase and orientation (Fig. 2). Each pair of oscillators α and β is characterized by the displacement vector $\mathbf{r} = \mathbf{x}_\beta - \mathbf{x}_\alpha$, the phase difference $\Delta\theta = \theta_\beta - \theta_\alpha$, and the oscillator angles χ_α and χ_β measured from the $\mathbf{e}_1$ direction (Fig. 2a). The two-dimensional pair correlation function $g(\mathbf{r})$ exhibits fluid-like structure with a single peak at $r = 25\,\mu\text{m} = 5a$, which is close to the maximum center-to-center separation $5.62a$ for particles arranged at the specified area fraction $\rho = 0.11$ (Fig. 2b). Despite the anisotropy of the oscillating particles, $g(\mathbf{r})$ is approximately isotropic except in the immediate vicinity of particle contacts as $r \to 2a$.

For neighboring particles separated by $r < 4a$, a joint histogram of phase and angle differences exhibits a sharp peak centered at the origin (Fig. 2c), indicating a strong tendency toward synchronization ($\Delta\theta \to 0$) and alignment ($\Delta\chi \to 0$). Additional peaks at the four corners of the plot are related to the central peak by phase-angle symmetry. The covariance of the peak is nearly isotropic (aspect ratio = 1.3), suggesting that synchronization and alignment arise from a common origin. By contrast, if these two forms of order arose from different mechanisms—such as hydrodynamic versus electrostatic interactions—their relative strengths would likely differ, producing an anisotropic peak elongated along one axis. To quantify this correlated synchronematic ordering, we define the order parameter $S_{\theta\chi} = \langle \cos(\Delta\theta)\cos(\Delta\chi)\rangle$, where the average is taken over oscillator pairs and over time. This parameter varies from zero for uncorrelated particles to one for perfect synchrony and alignment; the experimental data in Fig. 2c correspond to $S_{\theta\chi} = 0.57$.

The strength of phase-angle order decays exponentially with distance as $S_{\theta\chi} \propto e^{-r/\ell}$, defining a characteristic correlation length ℓ (Fig. 2d). For the experiment shown in Fig. 2a (Supplementary Video 1), we find $\ell = 35\,\mu\text{m} = 7.0a$, beyond which the oscillator phases and orientations become uncorrelated. These experimental results are reproduced by SD simulations (Fig. 2d; see also Supplementary Note 4), which provide further insights into the mechanism of local synchronization and alignment. Simulations in which electrostatic interactions are turned off yield the same correlation length, indicating that hydrodynamic interactions play the dominant role in establishing phase-angle order. We further observe that decreasing the rotational diffusivity D_χ from 5.0 to 3.5 rad^2/s results in a continuous increase in the correlation length ℓ, along with the emergence of a nonzero asymptotic value of $S_{\theta\chi}$ that reflects global correlations extending through the full simulation domain (see Supplementary Note 5).

The simulation results show that the correlation length ℓ emerges from the competition between hydrodynamic interactions and rotational diffusion. By balancing the rotational diffusivity D_χ due to field-driven oscillations and particle rotation in the disturbance flow of a neighboring oscillator, one obtains a simple estimate for this length scale. Briefly, a sphere rolling through a viscous fluid above a plane wall creates a far-field disturbance in the fluid vorticity that scales as $6aT/8\pi\eta r^4$, where η is the fluid viscosity, and T is the driving torque[45] (Supplementary Fig. S2). To first approximation, a neighboring sphere rotates at half this speed[46]. Using the standard model for Quincke rotation[27,28], the characteristic torque is

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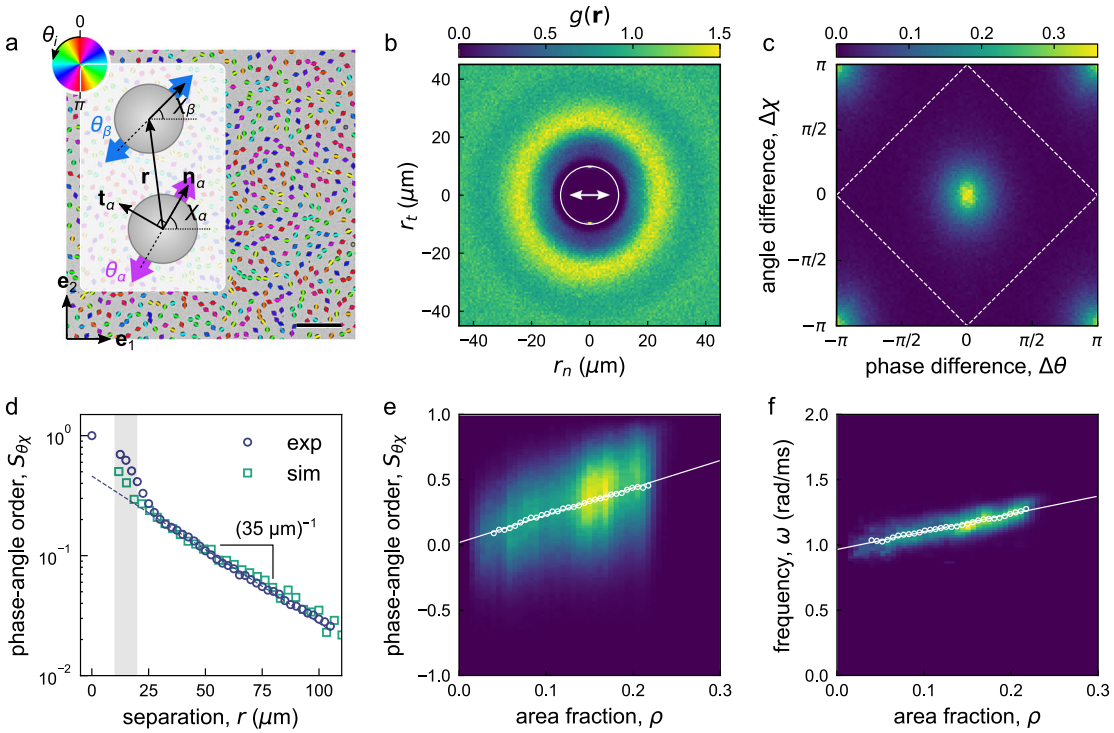


Fig. 2 | Fluid cluster. a Image of the fluid cluster with ellipsoidal markers denoting the orientation χ and phase θ of each oscillator. The inset shows a schematic of two oscillators highlighting relevant quantities. Scale bar is 100 μm. See Supplementary Video 1. **b** Pair correlation function $g(\mathbf{r})$ as a function of displacements parallel, $r_n = \mathbf{r}\cdot\mathbf{n}$, and perpendicular, $r_t = \mathbf{r}\cdot\mathbf{t}$, to the oscillation direction $\mathbf{n}$. **c** Time-averaged histogram of the angle difference, $\Delta\chi = \chi_\beta - \chi_\alpha$, and the phase difference, $\Delta\theta = \theta_\beta - \theta_\alpha$ for oscillator pairs α and β separated by one to two particle diameters. **d** Phase-angle order parameter, $S_{\theta\chi} = \langle\cos\Delta\theta\cos\Delta\chi\rangle$, as a function of separation r comparing experiments (circles) and simulations (squares). **e** For experiments with larger density variations in the imaging region (see Supplementary Video 3), the strength of phase-angle order (based on the nearest six neighbors) increases linearly with particle area fraction. The colormap shows a histogram over particles; markers denote the average order parameter $S_{\theta\chi}$. **f** The oscillation frequency ω also increases linearly with increasing area fraction.

$T = (8\pi\eta a^3/\tau_{\mathrm{mw}})[(E_e/E_c)^2 - 1]^{1/2}$, where $\tau_{\mathrm{mw}} \approx 0.7$ ms is the Maxwell–Wagner relaxation time, and $E_e/E_c \approx 6$ is the ratio between the external field strength and a critical field strength for spontaneous Quincke rotation[31]. The oscillation-averaged torque from the simulations is approximately 80% of this characteristic value. Balancing rotational diffusion and flow-induced rotation yields an estimate for the correlation length, $\ell \sim a(3E_e/D_\chi\tau_{\mathrm{mw}}E_c)^{1/4} \approx 9a$, in approximate agreement with experiment and simulation.

Additional experiments on fluid clusters with spatial density variations across the imaging region show increasing levels of synchronization, alignment, and frequency enhancement with decreasing interparticle separation (Supplementary Video 3). The phase-angle order parameter $S_{\theta\chi}$, computed based on each particle's six nearest neighbors, increases approximately linearly with area fraction up to $\rho \approx 0.2$ (Fig. 2e). These observations are attributed to the increasing strength of hydrodynamic interactions with decreasing particle separation. Notably, these interactions are non-reciprocal, enabling the mutual enhancement of the oscillation frequency among neighboring particles[39]. At low density, the average frequency increases linearly with area fraction, rising by ca. 30% over the range ρ = 0 to 0.2 (Fig. 2f).

Synchronematic crystals

To promote global synchronization, we initialize particle oscillations in locally dense regions prepared by activity-induced accumulation near fixed obstacles in the chamber (see "Methods"). The resulting clusters, which we refer to as synchronematic crystals, exhibit high values of the hexatic order parameter (Fig. 3a), which fluctuates about ~0.8 in the core. Within these self-oscillating assemblies, particles move back-and-forth along circular paths centered on a topological defect (Fig. 3b; Supplementary Video 4). While reminiscent of vortex structures formed by Quincke rollers[26,30], the present oscillations are achiral—consistent with the phase-angle symmetry of the constituent oscillators. Analysis of individual particles shows that they oscillate in synchrony at a common frequency and with nearly identical phases (Fig. 3c). The oscillator directors are locally aligned with neighboring particles, forming concentric rings centered on the defect.

SD simulations of particle clusters reproduce these experimental observations when augmented by attractive interactions that confine and stabilize the oscillator crystals. To model this confinement, we introduce a weak long-range attraction consistent with electroosmotic flows generated by particle-induced distortions of the electric field near the electrode surface (see "Methods")[40–44]. These extensile flows draw neighboring particles inward, counteracting the natural tendency of the oscillators to disperse. With this single modification, the simulations recover key features of the synchronematic crystals—namely, their crystalline order, synchronized oscillations, and circular alignment around a central defect (Supplementary Video 5). Moreover, by analyzing multiple clusters of varying sizes, we find quantitative agreement between simulation and experiment for the collective oscillation frequency as a function of cluster size (Fig. 3d).

Synchronematic crystals with circular ordering are distinguished by a central crystalline domain that rotates like a rigid body about a fluctuating origin. Relative to this origin, the average azimuthal velocity increases linearly with radial distance as $\langle|v_\theta|\rangle \approx \Omega r$, where $\Omega \approx 0.14$ rad/ms defines the average angular velocity of the cluster—consistent across different cluster sizes in both experiment and simulation (Fig. 3e). Due to the finite amplitude of particle oscillations—denoted A_{max}—this rigid-body-like motion can extend out to a radial distance $r_{\mathrm{max}} = \omega A_{\mathrm{max}}/\pi\Omega$. Beyond this threshold, the average azimuthal velocity reaches a maximum and then decays with increasing distance from the center. Experimental measurements on isolated

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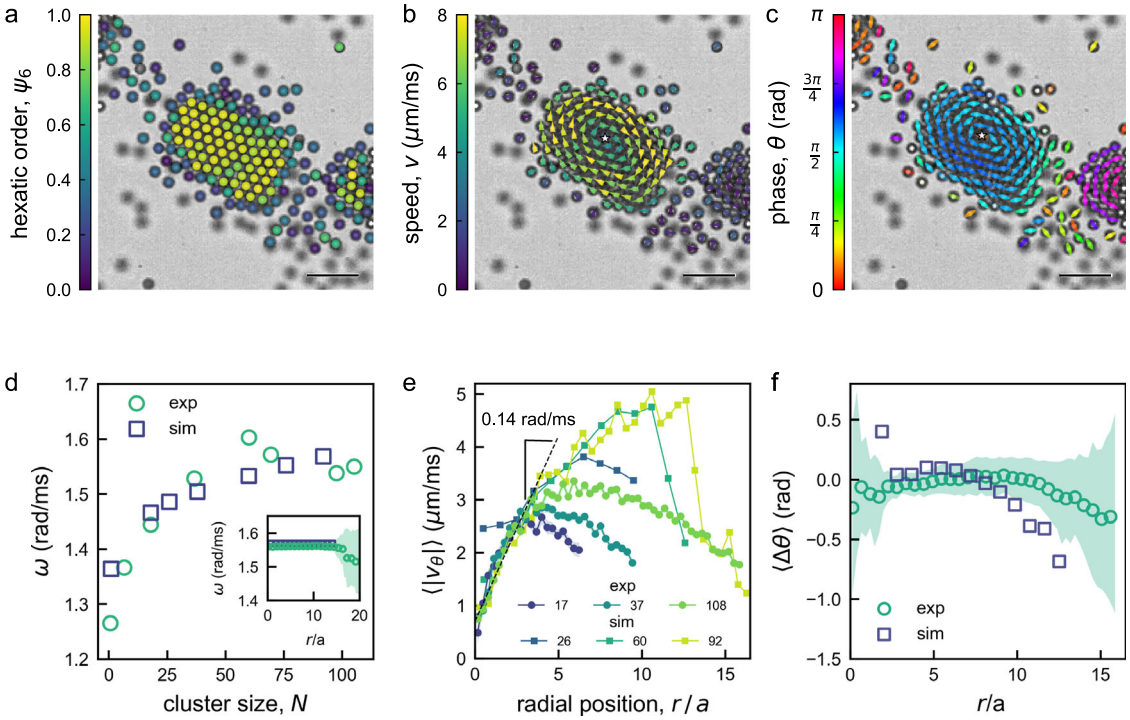


Fig. 3 | Synchronematic crystals. Image of a single particle cluster annotated to show: **a** colored disks representing the hexatic order parameter ψ_6; **b** colored triangles indicating the instantaneous particle velocity **v**; and **c** colored ellipsoids denoting the oscillation phase θ and orientation angle χ. Scale bars are 50 µm. See Supplementary Video 4. The star in (**b**, **c**) denotes the instantaneous center of cluster rotation and of circular alignment, respectively. **d** Oscillation frequency of synchronized particle clusters as a function of particle number N, comparing experiments (circles) and simulations (squares). The inset shows the oscillation frequency versus radial distance from the cluster center, illustrating synchronized motion within the cluster interior. **e** Time-averaged azimuthal particle speed $\langle|v_\theta|\rangle$ as a function of radial distance from the center of rotation, shown for experiments and simulations across different cluster sizes N. **f** Oscillation phase θ, measured relative to the cluster-averaged mean θ_c, plotted as a function of radial distance from the cluster center and averaged over multiple oscillation cycles: $\langle\Delta\theta\rangle = \langle\theta - \theta_c\rangle$. The shaded region denotes ±1 standard deviation.

particles suggest $A_{max} \approx 0.5a$ (Fig. 1b), which corresponds to a core radius $r_{max} \approx 3.5a$. By contrast, simulations indicate that $A_{max} \approx 1.0a$, resulting in higher particle speeds within simulated clusters. This discrepancy is attributed to differences in the strength of rotation-translation coupling between the particle and the nearby wall.

Like the particle velocity, the oscillation phase θ varies systematically with radial distance r from the cluster's central defect (Fig. 3f). Relative to the average phase in the synchronized assembly, θ increases with r from an ill-defined value near the origin, reaches a maximum at intermediate radii, and decreases near the outer edge. These spatial variations reflect a competition between reciprocal phase interactions that promote local synchronization and non-reciprocal interactions that enhance the oscillation frequency (see below)[39]. Particles near the periphery, with fewer interacting neighbors, tend to lag in phase relative to those in the bulk; however, the resulting phase gradients are modest and insufficient to disrupt synchronization for all cluster sizes studied (up to $N = 110$ particles).

Linear vs. circular ordering

In addition to circular ordering, Quincke oscillators can exhibit other transient modes of synchronematic order depending on the initial particle distribution. In elongated clusters, particles tend to align their synchronized oscillations along a common direction parallel to the cluster's principal axis (Fig. 4a). Over time, this linear oscillation mode becomes unstable, giving rise to two daughter clusters that adopt the favored circular ordering described above (Supplementary Video 6). To quantify this transition, we introduce two additional order parameters that measure the degree of linear and circular nematic order. The linear order parameter, $S_l = \frac{1}{2}\langle\cos(2(\chi - \chi_0)) + 1\rangle$, describes nematic alignment relative to a global angle χ_0 determined by the cluster's principal axis. The circular order parameter, $S_c = \frac{1}{2}\langle\cos(2(\chi - \mu - \frac{\pi}{2}))\rangle$, quantifies local alignment of oscillator directions around a central defect, where μ is the azimuthal angle of each particle's position relative to the cluster center. For the experiment shown in Fig. 4a, the transition from linear to circular ordering is marked by a sharp drop in S_l and a concomitant rise in S_c. The reverse transition is not observed, suggesting that linear ordering within elongated clusters is metastable and exists only as a transient.

Simulations reproduce the sharp transition from linear to circular ordering and reveal its underlying mechanisms. When particles are initialized within an elongated rectangular cluster with random phases and orientations, they rapidly synchronize and align along the cluster's principal axis. This linear synchronematic ordering remains stable for approximately 100 oscillation cycles before transitioning to circular ordering, as captured by the temporal evolution of the order parameters S_l and S_c (Fig. 4b). The transition is accompanied by a marked change in cluster geometry, quantified by the ellipticity λ, which increases from $\lambda \approx 0.4$ (elongated) to $\lambda \approx 0.65$ (circular). Together, these results confirm that linear ordering within elongated clusters is metastable and ultimately gives way to vortex-like oscillations within circular synchronematic crystals.

Additional simulations omitting either dipolar or hydrodynamic interactions show that both are essential for the sharp transition from linear to circular ordering (Fig. 4c). In the absence of hydrodynamics, electrostatic interactions alone—dominated by dipolar contributions—promote transient linear alignment; however, the cluster rapidly destabilizes and fragments. Conversely, without dipolar interactions, the cluster gradually adopts a circular shape, with hydrodynamic interactions alone driving the eventual onset of circular alignment. Only when both interactions are present does the system exhibit a

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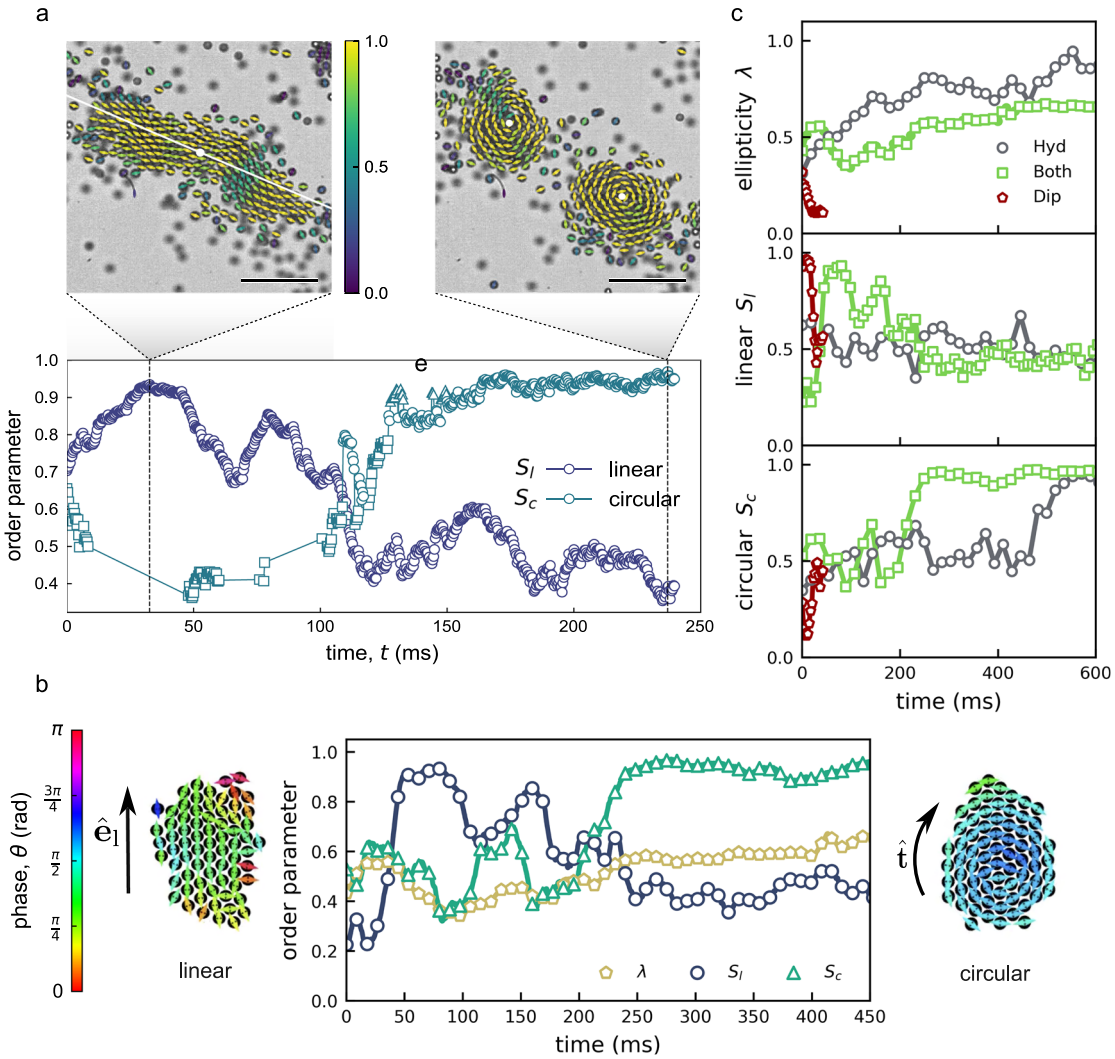


Fig. 4 | Transition from linear to circular synchronematic order. a Image of an elongated particle cluster with linear order (left) and the resulting daughter clusters with circular order (right). Particles are annotated by colored ellipsoids representing the oscillation direction and the strength of linear and circular order, respectively. The plot shows the time evolution of the respective order parameters S_l and S_c. Scale bar is 100 µm. See Supplementary Video 6. **b** Simulation results showing a similar transition from linear to circular ordering in a cluster of $N = 70$ particles. In addition to the order parameters S_l and S_c, the plot shows the transient ellipticity λ (reciprocal aspect ratio), highlighting the transition from elongated ($\hat{\mathbf{e}}_l$) to circular ($\hat{\mathbf{t}}$) orientation of the oscillations. See Supplementary Video 7. **c** Additional simulation results for the same cluster size, omitting either dipolar (purple) or hydrodynamic pair (red) interactions.

long-lived, elongated configuration with linear synchronematic order, followed by a sudden transition in both geometry and collective dynamics. These observations suggest that dipolar interactions favor alignment along the cluster's long axis, while hydrodynamic interactions drive the transition to vortex-like ordering.

Reduced-order swarmalator model for Quincke oscillators

To clarify the role of hydrodynamic interactions in directing synchronematic order, we derived a reduced-order model based on weakly coupled oscillator theory[47] that describes the evolution of oscillator phases and orientations. As summarized in the "Methods" and derived in Supplementary Note 6, we focus on the leading-order hydrodynamic interactions between widely separated oscillators, which scale as d^{-3} with oscillator separation d. For simplicity, we neglect the effects of electrostatic interactions at this order as they are comparatively weak and do not alter the types of phase-angle couplings present in the model (see Supplementary Fig. S10). Interactions that alter the oscillator positions are subdominant—scaling as d^{-4}—and therefore excluded. We use the adjoint, or phase-reduction, method[47,48] to compute the phase-response curve of the model limit cycle, which quantifies how small perturbations shift its oscillation phase. By projecting the far-field hydrodynamic interactions onto these response modes, we obtain coupled evolution equations for the oscillator phase and orientation that show how hydrodynamic coupling links synchronization and alignment in self-oscillating colloids.

The pairwise interaction functions, $H^{\theta}_{3h}(\psi_\alpha, \psi_\beta, \Delta\theta)$ and $H^{\chi}_{3h}(\psi_\alpha, \psi_\beta, \Delta\theta)$, describe how the hydrodynamic disturbance produced by oscillator β modifies the phase and orientation of oscillator α. These functions depend on the phase difference $\Delta\theta = \theta_\beta - \theta_\alpha$ and on the angles $\psi_\alpha = \chi_\alpha - \mu$ and $\psi_\beta = \chi_\beta - \mu$, which measure each oscillator's orientation relative to the line joining their centers. Here, μ denotes the azimuthal angle of the displacement vector from α to β in the plane of motion. An explicit low-order approximation to these interaction functions is provided in the "Methods" section, where their harmonic structure is derived from the symmetry of the far-field hydrodynamic flows. We note that both interaction functions display coupled dependencies: the phase interaction depends on the relative orientation between oscillators, and conversely, the angular interaction depends on their phase difference.

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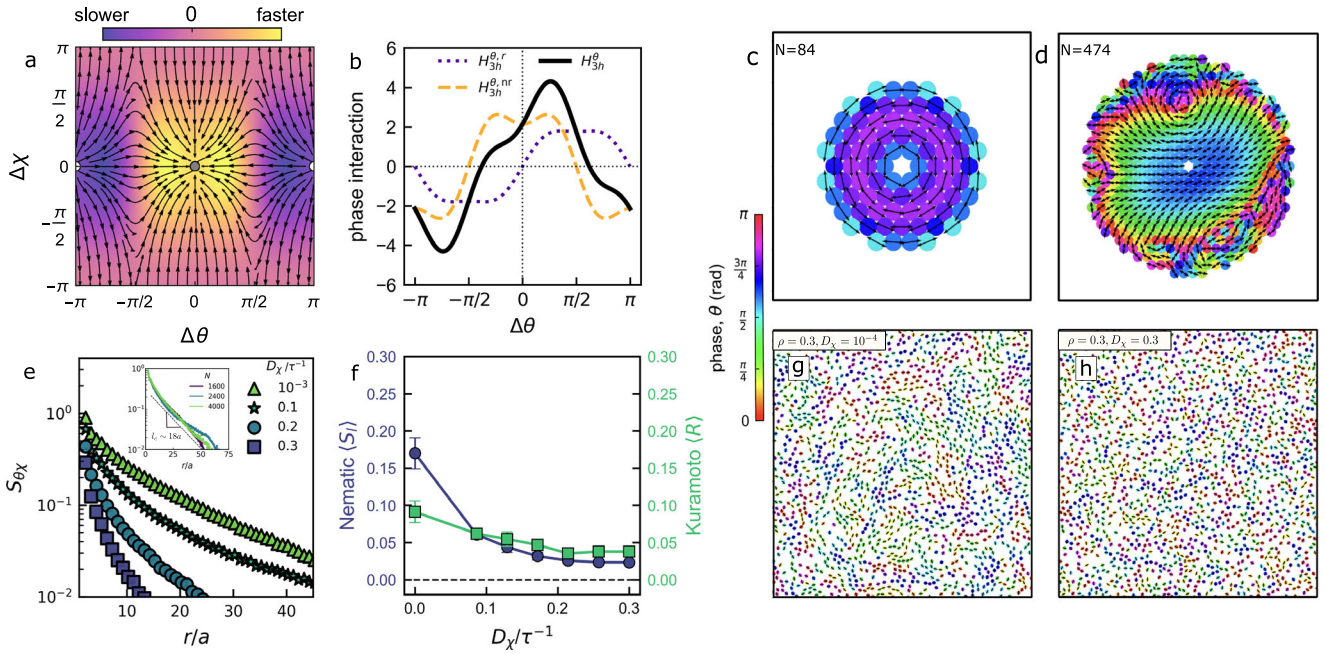


Fig. 5 | Reduced-ordered swarmalator simulations. a Phase portrait of two interacting oscillators with orientation angles constrained by $\psi_\alpha + \psi_\beta = 0$. Streamlines show the evolution of phase and orientation differences ($\Delta\theta$, $\Delta\chi$), while the colormap denotes the mean phase velocity $\frac{1}{2}(\theta_\alpha + \theta_\beta)$. Gray and white symbols mark stable and unstable fixed points, respectively. **b** Phase interaction function $H^{\theta}_{3h}(\Delta\theta, \psi_\alpha, \psi_\beta)$ for in-line oscillators ($\psi_\alpha = \psi_\beta = 0$), decomposed into its odd, reciprocal component $H^{\theta,r}_{3h}(\Delta\theta)$ (purple dotted) and even, non-reciprocal component $H^{\theta,nr}_{3h}(\Delta\theta)$ (orange dashed). Representative snapshots of reduced-order simulations showing **c** a small crystalline cluster ($N = 84$) with circular alignment and **d** a larger cluster ($N = 474$) where non-reciprocal interactions disrupt global order. **e** Spatial correlations of the synchronematic order $S_{\theta\chi}(r/a)$ at $\rho = 0.3$. The inset shows data for $D_\chi = 10^{-4}$ rad$^2/\tau$ and different system sizes N, which collapse in the low-noise regime, demonstrating a finite correlation length independent of system size. **f** Global alignment and synchronization quantified by the nematic $\langle S \rangle$ and Kuramoto $\langle R \rangle$ order parameters as functions of the rotational diffusivity D_χ. Simulation snapshots in the fluid phase at $\rho = 0.3$ for **g** $D_\chi = 10^{-4}$ rad$^2/\tau$ and **h** $D_\chi = 0.3$ rad$^2/\tau$. Here, $\tau = 0.44$ ms is the characteristic timescale used to non-dimensionalize the simulations (see "Methods").

The interaction functions reveal how hydrodynamic coupling drives not only synchronization and alignment but also mutual acceleration among interacting oscillators. These three effects are illustrated in Fig. 5a, which shows the phase portrait of two interacting oscillators with orientations satisfying $\psi_\alpha + \psi_\beta = 0$. The phase difference $\Delta\theta$ and angle difference $\Delta\chi$ evolve to a stable fixed point at the origin corresponding to synchronization and alignment along the line of centers. The colormap represents the mean phase velocity $\frac{1}{2}(\theta_\alpha + \theta_\beta)$, which increases near the synchronized configuration—revealing the mutual acceleration that accompanies synchronization. Such non-reciprocal effects are further illustrated in Fig. 5b, which decomposes the phase interaction function for in-line oscillators ($\psi_\alpha = \psi_\beta = 0$) into odd and even harmonics in the phase difference $\Delta\theta$. The odd components represent reciprocal, Kuramoto-like terms that promote phase locking[6], whereas the even components correspond to non-reciprocal couplings like those in the Sakaguchi-Kuramoto model[16].

Using the pairwise interaction functions, simulations of oscillator populations with fixed positions reproduce key experimental observations—namely, phase-locked circular alignment in finite crystalline clusters (Fig. 5c) and local synchronematic order in spatially disordered fluid phases (Fig. 5e–h; see also Supplementary Note 7). In crystalline clusters, the reduced model reproduces the phase-locked, circularly aligned states observed experimentally and in full SD simulations (Fig. 3). The collective oscillation frequency increases with cluster size due to non-reciprocal phase interactions, eventually saturating for larger clusters (Supplementary Fig. S12). Within the cluster, the oscillation phase exhibits radial gradients: it is retarded near the center and outer edge and advanced in an intermediate annular region (Supplementary Video 9). Beyond a critical size, these phase gradients grow to disrupt global phase-locking and circular alignment in favor of local synchronization and linear alignment (see Fig. 5d, Supplementary Video 10). Though not observed experimentally, this result suggests a mechanism by which hydrodynamic interactions might limit the size of synchronematic assemblies.

Large-scale simulations of oscillators randomly distributed on a periodic domain exhibit local synchronematic order with a finite correlation length that depends on the rotational diffusivity D_χ (Fig. 5e–h). Neighboring oscillators tend to adopt a common phase and orientation, as characterized by the synchronematic order parameter $S_{\theta\chi}(d)$, which decays exponentially with oscillator separation d (Fig. 5e). Despite qualitative agreement with experiments and the full SD simulations, the reduced model—based on far-field hydrodynamic couplings—tends to overestimate the interaction strength at short range. As a result, higher levels of rotational noise are required to disrupt synchronematic order and reproduce the experimentally observed correlation length. Notably, even in the absence of noise, the system fails to exhibit global synchronization and alignment due to non-reciprocal contributions in the phase interactions (Fig. 5f; see also Supplementary Video 11). These contributions lead to cooperative acceleration among nearby oscillators, which amplifies the effects of local density fluctuations and induces spatial variations in oscillation frequency—ultimately frustrating the emergence of quasi-long-range order. Consistent with this interpretation, simulations on a uniform lattice—without positional disorder—exhibit longer range phase and orientational order. Similar quasi-long-range correlations are realized in disordered systems by removing the non-reciprocal contributions from the phase interaction functions (Supplementary Fig. S13).

Our synchronematic model draws conceptual connections to both the Kuramoto and XY models but introduces key distinctions through its coupling of phase and orientation. At low temperature, the XY model exhibits quasi-long-range order maintained by bound vortex-antivortex pairs—topological defects with opposite winding that suppress large-scale disorder. Similar features emerge in the

synchronematic model on a square lattice: simulations at low rotational noise show power-law decay in both orientational and phase correlations, together with the formation of corresponding defects, including analogs of vortices and disclinations (Supplementary Fig. S13 and Supplementary Video 12). Distinct types of defects appear to coexist—some primarily associated with phase, others with orientation—revealing a rich coupling between the two fields and pointing to broader connections with the classification of topological excitations in active matter[49]. The nature of these coupled phase-angle defects, their dynamics and interactions, and the ordering transitions they undergo with increasing noise remain poorly understood—highlighting open questions for future theoretical and computational work.

Additionally, unlike XY and Kuramoto models, the synchronematic model includes interactions that cannot be derived from a scalar potential, further enriching the system's dynamical behaviors (Supplementary Note 8). Such non-conservative interactions enable other types of spatio-temporal order within assemblies of synchronematic colloids. For example, simulations of linear chains predict spontaneous alignment and phase-locking into stable metachronal waves—traveling oscillatory patterns that propagate along the length of the chain (Supplementary Fig. S14). Similar wave-like behaviors have been reported in simple models of ciliary beating, where hydrodynamic interactions generate non-reciprocal phase couplings analogous to those described here[50].

Discussion

Populations of Quincke oscillators reveal how phase synchronization and orientational alignment can co-emerge through hydrodynamic coupling, giving rise to a new class of active order that integrates the physics of coupled oscillators and active matter. The circular oscillations of synchronematic crystals are strikingly similar to those of active solids formed by bacteria embedded in elastic matrices[51]; however, their oscillatory dynamics are intrinsic to the active units rather than emergent from their interactions with an elastic medium[52]. Whereas polarity or chirality often underlie activity-induced order[3,14,53], synchronematic order arises in an achiral, nematic system governed by hydrodynamic interactions. Its collective dynamics are further shaped by geometry: confinement and lattice topology modulate spatial correlations and promote propagating modes reminiscent of metachronal waves in ciliary carpets[5,7].

Building on these findings, the pursuit of active oscillatory materials will benefit from biological inspiration such as that provided by the dynamic instability of microtubules and the multicellular morphogenesis of myxobacteria. The cytoskeleton uses dissipative processes to break detailed balance and direct the assembly of structures that are simultaneously rigid and dynamically reconfigurable. By analogy, synchronematic crystals of Quincke oscillators exhibit catastrophic disassembly events followed by longer periods of particle recruitment (Supplementary Video 8), suggesting opportunities for temporal control in active materials. The oscillatory mobility of rod-shaped myxobacteria further illustrates how the coupling of cell orientation with oscillation phase can enable populations to coordinate motion and organize multicellular patterns[54].

With greater control over oscillator geometry and configuration, future work on self-oscillating colloids can explore new modes of synchronematic order and probe the effective interactions among oscillators and between oscillator clusters. These interactions—especially the hydrodynamic couplings that mediate alignment and synchronization—may be tuned through particle shape, size, or polydispersity to engineer collective responses. Ultimately, such principles may guide the design of colloidal materials in which self-oscillating components organize in space and time to realize programmable functions.

Methods

Experiment

Poly(polystyrene) spheres with radius a = 5 μm are dispersed in a 0.15 M solution of AOT surfactant in hexadecane at a volume fraction of ~0.01. The suspension is injected into a microfluidic chamber sandwiched between two ITO-coated glass electrodes (Sigma-Aldrich, 70–100 Ω/sq) separated by 150 μm. A source measure unit (Keithley 2410-C) supplies a constant voltage of 900 V between the electrodes to produce an electric field of strength E_e = 6 V/μm. These conditions—particle size, AOT concentration, and field strength—were previously shown to produce Quincke oscillations[31]. Particle motion at the lower electrode is imaged from below using an optical microscope under brightfield illumination. Videos are captured at 3000 frames per second by a high-speed camera and analyzed to detect particles and link trajectories using TrackPy (v0.6.1).

To promote the formation of fluid or crystalline clusters, we use one of two different methods to prepare the initial particle distributions prior to activating Quincke oscillations. In the first method, a DC electric field of strength E_e = 2 V/μm is first applied for 30 s to induce Quincke rolling, which randomizes particle positions across the chamber. The field strength is then lowered to 1 V/μm for 30 s to halt rolling and separate nearby particles via repulsive dipolar interactions. The resulting particle distribution is approximately uniform, aside from long-wavelength density fluctuations attributed to weak field-induced flows. In the second method, we apply pieces of Scotch tape to the lower electrode to create large (a few mm) obstacles that obstruct the motion of the Quincke rollers. When the electric field of strength E_e = 2 V/μm is applied for 5 s, rolling particles accumulate near these obstacles to create local regions of high particle density. In both methods, the overall area fraction of particles remains similar. Control over the initial particle distribution remains limited, motivating future improvements in preparation methods.

Data analysis

We analyzed each particle trajectory $\mathbf{y}(t)$ using Bayesian inference to extract latent variables characterizing the underlying oscillator: the position $\mathbf{x}(t)$, relative phase $\phi(t)$, and orientation angle $\chi(t)$, all of which were modeled as independent diffusive processes (see Supplementary Note 1 for details). The observed position was related to these variables via the generative model $\mathbf{y} = \mathbf{x} + \mathbf{n}(\chi)g(\omega t + \phi)$, where $\mathbf{n}$ is a unit vector along the oscillation axis, and $g(\theta)$ is a 2π-periodic waveform approximated by a truncated Fourier series. Model parameters—including the oscillation frequency, phase, and angle diffusivities, and waveform shape—were estimated along with the latent variables using the expectation-maximization (EM) algorithm.[55] Each EM iteration alternated between estimating the latent trajectories via an extended Kalman filter (EKF)[55–57] and updating model parameters to maximize the expected log-likelihood. After fitting the model to a small subset of trajectories in each experiment, we fixed the parameters and used the EKF alone to infer the latent variables for all remaining particles.

Stokesian Dynamics simulations

We simulate collections of oscillating colloids above a plane wall using the SD method[35,36] implemented in an open-source codebase developed by Donev and collaborators[37]. This framework resolves many-body hydrodynamic interactions among rotating and translating spheres via the grand mobility matrix $\mathcal{M}(\mathbf{q})$, and governs the evolution of particle positions and orientations through:

$$\frac{d\mathbf{q}}{dt} = \mathcal{M}\cdot\mathcal{F} + (k_B T)\nabla_{\mathbf{q}}\cdot\mathcal{M} + \sqrt{2k_B T}\,\mathcal{M}^{1/2}\boldsymbol{\xi}(t) \tag{1}$$

where $\mathbf{q}$ is the generalized coordinate vector containing the positions and orientations of the N particles, $\mathcal{F}$ is the generalized force and torque vector, $k_B T$ is the thermal energy, and $\boldsymbol{\xi}(t)$ is a vector of Gaussian

white noise. The mobility tensor $\mathcal{M}$ captures near-field lubrication interactions and far-field hydrodynamics via the Rotne-Prager-Yamakawa-Blake tensor[35,36].

Forces and torques. The force-torque vector $\mathcal{F}$ includes contributions due to (1) electrostatic forces and torques, (2) frictional particle-wall interactions, and (3) excluded-volume interactions among the particles. The electric force and torque on particle i at position $\mathbf{r}_i$ with charge q_i, dipole $\mathbf{p}_i$, and quadrupole $\mathbf{Q}_i$ are given by

$$\begin{aligned}\mathbf{F}_i^{\mathrm{e}} &= 4\pi\varepsilon_{\mathrm{f}}\left(q_i\mathbf{E}(\mathbf{r}_i)+\mathbf{p}_i\cdot\nabla\mathbf{E}(\mathbf{r}_i)+\frac{1}{6}\mathbf{Q}_i[\cdot]^3\nabla\nabla\mathbf{E}(\mathbf{r}_i)+\ldots\right)\\ \mathbf{T}_i^{\mathrm{e}} &= 4\pi\varepsilon_{\mathrm{f}}\left(\mathbf{p}_i\times\mathbf{E}(\mathbf{r}_i)+\frac{1}{3}(\mathbf{Q}_i\cdot\nabla)\times\mathbf{E}(\mathbf{r}_i)+\ldots\right)\end{aligned} \tag{2}$$

Here, the electric field $\mathbf{E}$ includes contributions from the external field $\mathbf{E}_{\mathrm{e}} = E_{\mathrm{e}}\mathbf{e}_3$ directed normal to the substrate (3-direction) as well as the disturbance field $\mathbf{E}_{\mathrm{d}}$ due to neighboring particles. In our SD simulations, we neglect the effects of the nearby electrode when computing the disturbance fields (i.e., no image charges, dipoles, or quadrupoles; see Supplementary Note 2 for a discussion of their contributions). The resulting electrostatic interactions between (unscreened) charge monopoles are strongly repulsive, resulting in nonphysical particle dynamics. We therefore exclude the contribution of the monopole in the force (2) and in the electric disturbance field $\mathbf{E}_{\mathrm{d}}$, which becomes

$$\mathbf{E}_{\mathrm{d}}(\mathbf{r}_i)=\sum_{j\neq i}\left(\frac{1}{r_{ij}^3}(3\hat{\mathbf{r}}_{ij}\hat{\mathbf{r}}_{ij}-\boldsymbol{\delta})\cdot\mathbf{p}_j+\frac{1}{2r_{ij}^4}(5\hat{\mathbf{r}}_{ij}\hat{\mathbf{r}}_{ij}\hat{\mathbf{r}}_{ij}-2\boldsymbol{\delta}\hat{\mathbf{r}}_{ij}):\mathbf{Q}_j\right) \tag{3}$$

where $\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$ is the displacement from particle j to particle i with magnitude r_{ij} and direction $\hat{\mathbf{r}}_{ij}$. For simplicity, we neglect electrostatic interactions that decay faster than r^{-4}; the retained contributions include dipole-dipole interactions in the force and torque as well as dipole-quadrupole interactions in the torque.

At the single particle level, for reproducing the amplitudes and frequencies of the oscillations observed experimentally, we introduce an additional friction force $\mathbf{F}_{\mathrm{f}}$ and torque $\mathbf{T}_{\mathrm{f}}$ that ensures the following kinematic condition above the wall plane: $\mathbf{U} = \lambda a\boldsymbol{\Omega}\times\mathbf{e}_3$, where $\boldsymbol{\Omega}$ the angular velocity of the particle, and λ a dimensionless factor that characterizes the strength of the rotation-translation coupling. The value $\lambda = 0.6$ reproduces the rolling speeds and oscillations observed in experiments for a single Quincke oscillator. As a result, this additional friction force and torque are proportional to the electrostatic torque and force driving the particle motion (see Supplementary Note 2).

Finally, we model excluded-volume interactions between particles of radius a using a capped Yukawa (screened-exponential) repulsion.

$$U(r)=\begin{cases}U_0\exp\left[-(r-2a)/\lambda_D\right], r\geq 2a,\\ U_0\left[1+\frac{2a-r}{\lambda_D}\right], r<2a,\end{cases} \tag{4}$$

Here, U_0 sets the interaction strength, λ_D is the decay length. The piecewise definition ensures that $U(r)$ is continuous at $r = 2a$, while capping the force at short distances.

Moment dynamics. The time evolution of the particles' charge, dipole, and quadrupole moments is governed by equations previously derived from the leaky-dielectric model[58] extended to include gradients in electrolyte conductivity[39]

$$\frac{dq_i}{dt}=-\frac{1}{\tau}q-\frac{\boldsymbol{\gamma}}{\tau}\cdot\left(\frac{2}{3a}\mathbf{p}+\frac{a^2}{3}\mathbf{E}\right) \tag{5}$$

$$\frac{d\mathbf{p}_i}{dt}=\boldsymbol{\Omega}\times\left(\mathbf{p}-a^3\varepsilon'_{\mathrm{cm}}\mathbf{E}\right)-\frac{1}{\tau'}\left(\mathbf{p}-a^3\kappa'_{\mathrm{cm}}\mathbf{E}\right)-\frac{\boldsymbol{\gamma}'}{\tau'}\cdot\left(aq\boldsymbol{\delta}+\frac{3}{5a}\mathbf{Q}\right) \tag{6}$$

$$\frac{d\mathbf{Q}_i}{dt}=2[\boldsymbol{\Omega}\times\mathbf{Q}]^{\mathrm{st}}-\frac{1}{\tau''}\mathbf{Q}-\frac{1}{\tau''}\left[\boldsymbol{\gamma}''\left(4a\mathbf{p}+2a^4\mathbf{E}\right)\right]^{\mathrm{st}} \tag{7}$$

As detailed in Supplementary Note 2, the time scales τ, τ', and τ'' represent the relaxation times for the charge, dipole, and quadrupole, respectively. The dimensionless gradient parameters $\boldsymbol{\gamma}$, $\boldsymbol{\gamma}'$, and $\boldsymbol{\gamma}''$ are each proportional to the conductivity gradient $\nabla\kappa_{\mathrm{f}}$ and act to couple the dynamics of the respective moments. The Claussius–Mossotti factors $\varepsilon'_{\mathrm{cm}}$ and κ'_{cm} for the dipole represent the high- and low-frequency polarizabilities, respectively. The superscript "st" denotes the symmetric and traceless part of the second-order tensor. These dynamics neglect additional contributions due to gradients in the electric field (see Supplementary Note 2).

Athermal noise. In experiments, the measured orientational diffusivity D_χ of the oscillators is orders of magnitude larger than the value expected from purely Brownian motion. Setting the thermal energy $k_{\mathrm{B}}T$ to an effective value that reproduces the experimental D_χ would, however, lead to translational fluctuations far larger than those observed. We therefore neglect the effect of thermal noise by setting $k_{\mathrm{B}}T = 0$ in Eq. (1), and instead introduce an athermal white noise acting on the orientation of the dipole in Eq. (6). This athermal noise is implemented by drawing a random angular velocity from a Gaussian distribution, which changes the direction of the dipolar moment while keeping its magnitude constant. The orientation of the dipole χ_i thus evolves as

$$\dot{\chi}_i=\sqrt{2D_\chi}\,\xi_i(t), \tag{8}$$

where $\xi_i(t)$ is a Gaussian white noise with zero-mean and unit variance, satisfying $\langle\xi_i(t)\,\xi_j(t')\rangle = \delta_{ij}\,\delta(t-t')$. At each time step, the dipolar moment is updated by multiplying it by the rotation vector $(\cos\chi_i,\ \sin\chi_i)$, which alters its orientation without modifying its magnitude. This stochastic term, therefore, introduces random reorientations of the dipolar axis—coinciding with the instantaneous rotation axis of the oscillator—while preserving the instantaneous dipole strength.

Long-ranged attraction in crystalline clusters. In experiments, we see evidence of long-range attraction between particles within isolated dense clusters. Attraction in similar experimental setups was reported previously[40–44], and attributed to electroosmotic flows created by the distortion of the electric field introduced by the particles. To simulate this effect, we introduce an additional component to the velocity of particle i due to the flow generated by particle j of the form

$$\mathbf{U}_i^{\mathrm{a}}=-\frac{k\,e^{-r_{ij}/\ell_s}}{r_{ij}^2}\hat{\mathbf{r}}_{ij} \tag{9}$$

where k is the strength of the attraction, and the screening length ℓ_s describes the attenuation of the electroosmotic flow. The exponential screening in Eq. (9) regularizes the long-ranged r^{-2} interaction for computational tractability, allowing a finite cutoff ($\ell_s \sim 10a$) while approximating the screened electroosmotic flows predicted by theory[41].

Model units and simulation parameters. The above equations can be made dimensionless by means of setting the radius of the particle a as the unit length, and also setting the fluid permittivity ε_{f}, conductivity κ_{f}, and viscosity η to 1. In these units, time is scaled by $\tau = \varepsilon_{\mathrm{f}}/\kappa_{\mathrm{f}} = 0.44$ ms, and the electric field by $E_{\mathrm{s}} = (\eta\kappa_{\mathrm{f}}/\varepsilon_{\mathrm{f}}^2)^{1/2} = 0.60$ V/μm. Particle charges, dipoles, and quadrupoles are scaled by a^2E_{s}, a^3E_{s}, and a^4E_{s}, respectively. For a single oscillator, the SD simulations agree well with previous measurements[31] of the field-dependent rolling speed and oscillation frequency (see Supplementary Note 2). In these natural

units, the remaining dimensionless parameters are selected to match the experimental system: particle permittivity $\varepsilon_p = 1.2$, particle conductivity $\kappa_p = 0$, and external field strength $E_e = 10$. For the fluid cluster simulations shown in Fig. 2d, we use a rotational diffusion coefficient of $D_\chi = 5\ \mathrm{rad}^2/\mathrm{s}$ and an area fraction of $\rho = 0.12$, For the synchronematic crystal simulations shown in Fig. 3d–f, we use the same parameters but introduce the pairwise attraction of Eq. (9) with strength $k = 1.6$ and range $\ell_s = 10$. The excluded-volume interaction parameters are $\lambda_D = 4 \times 10^{-2}$ and $U_0 = 70$ in all simulations.

Numerical integration. We integrate Eq. (1) using the Stochastic Trapezoidal Split method with a fixed time step of $\Delta t = 0.1$, which provides second-order accuracy. The charge, dipole, and quadrupole Eqs. (5)–(7) are integrated using a fourth-order Runge–Kutta method with a smaller substep $\Delta t' = \Delta t/10$. This approach resolves the multipole dynamics ten times within each SD step, improving numerical accuracy. For the fluid-phase simulations shown in Fig. 2d, we simulate $N = 400$ particles under periodic boundary conditions and average results over five random initializations. Simulations without periodic boundaries yield qualitatively similar behavior. The simulations of finite crystalline clusters shown in Fig. 3d–f use an unbounded domain in place of periodic boundaries.

Swarmalator simulations

We simulate a collection of N oscillators indexed by $\alpha = 1 \ldots N$, each characterized by its fixed position $\mathbf{x}_\alpha$, internal phase θ_α, and in-plane orientation χ_α. For weakly coupled oscillators with stationary positions, the phase and orientation evolve through pairwise interactions and Gaussian noise according to

$$\dot{\theta}_\alpha = \omega + \sum_{\beta \neq \alpha} \frac{1}{d^3} H_3^\theta(\psi_\alpha, \psi_\beta, \Delta\theta) + \sqrt{2D_\chi}\, \xi_\alpha^\chi(t), \tag{10}$$

$$\dot{\chi}_\alpha = \sum_{\beta \neq \alpha} \frac{1}{d^3} H_3^\chi(\psi_\alpha, \psi_\beta, \Delta\theta) + \sqrt{2D_\theta}\, \xi_\alpha^\theta(t), \tag{11}$$

where ω is the oscillation frequency, $d = |\mathbf{x}_\beta - \mathbf{x}_\alpha|$ is the oscillator separation, and $\Delta\theta = \theta_\beta - \theta_\alpha$ is the phase difference. The angles $\psi_\alpha = \chi_\alpha - \mu$ and $\psi_\beta = \chi_\beta - \mu$ are measured relative to the line of centers connecting particles α and β, which has orientation μ. The interaction functions $H_3^\theta(\psi_\alpha, \psi_\beta, \Delta\theta)$ and $H_3^\chi(\psi_\alpha, \psi_\beta, \Delta\theta)$ are derived from our physical model in the limit of large separations $d \gg 1$ (see Supplementary Note 7). The Gaussian noise terms ξ_α^θ and ξ_α^χ are independent, zero-mean, unit-variance white noises, scaled by the noise strengths D_θ and D_χ, respectively. Equations (10) and (11) define a "swarmalator" model that couples phase synchronization with orientational alignment.

Hydrodynamic interaction functions. The far-field hydrodynamic coupling between oscillators is represented by the interaction functions $H_{3\mathrm{h}}^\theta(\psi_\alpha, \psi_\beta, \Delta\theta)$ and $H_{3\mathrm{h}}^\chi(\psi_\alpha, \psi_\beta, \Delta\theta)$, which are accurately approximated by the truncated Fourier series

$$H_{3\mathrm{h}}^\theta(\psi_\alpha, \psi_\beta, \Delta\theta) = (A_1^\theta \cos\Delta\theta + B_1^\theta \sin\Delta\theta + A_3^\theta \cos 3\Delta\theta + B_3^\theta \sin 3\Delta\theta)\, \cos\psi_\alpha \cos\psi_\beta \tag{12a}$$

$$H_{3\mathrm{h}}^\chi(\psi_\alpha, \psi_\beta, \Delta\theta) = (A_1^\chi \cos\Delta\theta + B_1^\chi \sin\Delta\theta + A_3^\chi \cos 3\Delta\theta + B_3^\chi \sin 3\Delta\theta)\, \sin\psi_\alpha \cos\psi_\beta \tag{12b}$$

The coefficients are estimated by linear regression of the computed interaction functions on a regular grid in ψ_α, ψ_β, $\Delta\theta$ space: $A_1^\theta = 2.921$, $B_1^\theta = 2.070$, $A_3^\theta = -0.781$, $B_3^\theta = 0.312$, $A_1^\chi = -5.351$, $B_1^\chi = 0.223$, $A_3^\chi = 3.009$, $B_3^\chi = -1.359$. All quantities are reported using the scaled variables and parameter estimates detailed in the previous section. This parametric approximation, neglecting higher order terms, is accurate to ca. 6%.

Numerical integration. Equations (10) and (11) are integrated using an explicit Euler–Maruyama scheme with time step $\Delta t = 0.1\tau$, where $\tau = 0.44$ ms is the same dimensionless time used for SD simulations. At each step, the phases and orientations of all oscillators are updated according to the deterministic drift terms and Gaussian random increments with variance $2D\Delta t$ in the corresponding channel. Reducing Δt by a factor of two produces no change in the measured observables, indicating numerical convergence. In all simulations, we set the phase diffusivity to zero, $D_\theta = 0$, such that noise is introduced only through the rotational diffusion of the oscillators. Each simulation condition is averaged over five independent realizations to estimate statistical variability. For larger systems ($N > 1600$), we employ a cell list with a cutoff length $l_c = 50a$ to efficiently reach the steady state. This cell size provides a good balance between computational cost and accuracy, yielding statistical properties indistinguishable from those obtained with smaller cell sizes.

We consider two types of simulations: finite clusters (Fig. 5c, d) and periodic domains (Fig. 5g, h). To mimic the synchronematic crystal, we arrange N oscillators on a triangular lattice inside a circular region of radius R, removing the central site to reproduce the immobile core observed experimentally. Initial phases and orientations are drawn independently from a uniform distribution on $[0, 2\pi)$, and the system is evolved in time until reaching a statistically stationary state. To model the fluid phase, we distribute N non-overlapping oscillators at random positions within a square domain with periodic boundary conditions, again initializing the phases and angles uniformly on $[0, 2\pi)$. The dynamics are then integrated in time until the system reaches steady-state statistics.

Order parameters. Simulations of the oscillator fluid are used to probe spontaneous pattern formation, the coexistence of synchronized domains, and the dynamics of topological defects. To quantify global phase synchronization, we compute the Kuramoto order parameter R as

$$Re^{i\Theta} = \frac{1}{N} \sum_{\alpha=1}^{N} e^{i\theta_\alpha} \tag{13}$$

where Θ is the mean phase. Similarly, to measure global orientational alignment, we calculate the nematic order parameter S as

$$Se^{i2\Psi} = \frac{1}{N} \sum_{\alpha=1}^{N} e^{i2\chi_\alpha} \tag{14}$$

where Ψ is the mean angle. Additionally, we monitor the average phase velocity of the oscillators, $\Omega = \langle N^{-1} \sum_\alpha \dot{\theta}_\alpha \rangle_t$, as well as the presence of topological defects in the orientation and phase fields.

Data availability

The microscopy videos generated and analyzed in this study are available on figshare at https://doi.org/10.6084/m9.figshare.30946217. Source data are provided with this paper.

Code availability

Swarmalator simulation code is available on GitHub at https://github.com/slevinskygra/ReducedSynchronematic.

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Author contributions

K.J.M.B. conceived the project. Z.Z. performed the experiments and analyzed the experimental data. S.G.L. developed and performed the Stokesian Dynamics simulations. K.J.M.B. and S.G.L. developed the reduced-order model and performed the swarmalator simulations. All authors contributed to data interpretation. S.G.L. drafted the manuscript with contributions from K.J.M.B. and M.O.d.l.C. K.J.M.B. and M.O.d.l.C. supervised the research and secured funding. All authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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